

YEREVAN ZVARTNOTS, Armenia

WMO: 377880

Lat: 40.150N

Lon: 44.383E

Elev: 854

StdP: 91.48

Time Zone: 4.00 (E04)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.2	-13.7	-19.1	0.8	-16.4	-16.0	1.0	-12.0	7.8	7.1	5.5	5.6	0.4	350	0.477

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.3	37.2	19.0	35.8	19.1	34.1	18.9	21.7	33.1	20.6	31.9	19.7	31.0	2.3	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.8	14.2	28.4	16.8	13.2	26.4	15.8	12.4	25.4	67.8	33.4	63.7	31.9	60.4	31.2	25.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	9.1	7.7	6.2	DB	-18.5	39.5	6.0	1.4	-22.8	40.6	-26.3	41.4	-29.6	42.2	-34.0	43.2
(5)				WB	-18.8	23.4	5.8	1.2	-23.0	24.2	-26.4	24.9	-29.6	25.5	-33.9	26.4

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	13.0	-3.3	0.8	7.4	13.0	18.1	23.3	27.0	26.9	21.5	14.4	6.3	-0.5	(6)
(7)		DBStd	10.99	5.55	5.08	4.01	3.65	2.96	3.01	2.82	2.67	3.32	3.25	3.64	5.09	(7)
(8)		HDD10.0	1229	414	257	98	12	0	0	0	0	0	5	117	326	(8)
(9)		HDD18.3	2783	672	491	338	164	40	2	0	0	9	124	360	584	(9)
(10)		CDD10.0	2318	0	0	19	102	252	400	527	522	346	143	7	0	(10)
(11)		CDD18.3	829	0	0	0	3	33	152	268	264	106	4	0	0	(11)
(12)		CDH23.3	10163	0	0	2	63	372	1747	3353	3326	1221	78	0	0	(12)
(13)		CDH26.7	4880	0	0	0	13	81	777	1769	1740	491	9	0	0	(13)
(14)	Wind	WSAvg	1.8	0.7	1.0	1.9	2.1	2.0	2.6	3.2	3.1	2.2	1.3	1.0	0.7	(14)
(15)	Precipitation	PrecAvg	281	23	22	31	37	43	20	12	8	9	29	25	23	(15)
(16)		PrecMax	470	61	102	82	103	114	94	47	50	93	91	76	83	(16)
(17)		PrecMin	116	0	2	0	10	10	0	0	0	0	0	0	0	(17)
(18)		PrecStd	87	17	19	22	23	26	19	11	11	17	22	22	17	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.2	15.0	22.7	28.2	31.0	36.9	39.7	39.2	36.1	28.0	19.4	11.8	(19)
(20)			MCWB	5.0	7.6	11.2	14.3	15.6	18.1	18.7	19.0	17.7	15.3	11.5	7.0	(20)
(21)		2%	DB	8.0	12.8	19.1	25.0	28.9	34.7	37.5	37.2	33.2	25.2	16.8	9.1	(21)
(22)			MCWB	4.2	6.3	9.8	12.7	15.4	17.8	19.2	18.9	17.5	14.2	9.8	5.6	(22)
(23)		5%	DB	5.9	10.6	17.0	22.6	27.1	32.9	35.9	35.8	31.2	23.8	14.9	7.2	(23)
(24)			MCWB	2.9	5.5	8.6	11.9	14.9	17.6	19.5	19.2	17.1	13.8	9.0	4.7	(24)
(25)		10%	DB	4.0	8.6	14.9	20.4	25.2	31.1	34.1	34.1	29.8	21.9	13.0	5.9	(25)
(26)			MCWB	1.8	4.4	7.9	11.3	14.4	17.4	19.4	19.1	16.8	13.3	8.2	3.9	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.5	8.9	12.1	15.3	17.9	20.7	23.1	23.4	20.4	16.7	12.5	7.7	(27)
(28)			MCDWB	8.8	13.4	20.0	24.3	27.2	31.9	34.5	34.6	31.6	24.1	17.0	10.4	(28)
(29)		2%	WB	4.6	7.2	10.6	13.8	16.8	19.4	21.8	21.8	19.0	15.5	11.0	6.3	(29)
(30)			MCDWB	6.8	11.3	17.2	21.3	25.0	30.6	33.3	33.2	30.3	22.1	14.7	8.0	(30)
(31)		5%	WB	3.2	5.8	9.4	13.0	16.0	18.7	20.7	20.6	18.0	14.6	9.8	5.2	(31)
(32)			MCDWB	5.4	9.7	15.3	19.9	24.2	29.7	32.4	31.7	29.1	21.2	13.5	6.6	(32)
(33)		10%	WB	1.8	4.6	8.4	12.1	15.3	17.9	20.0	19.7	17.1	13.9	8.7	4.0	(33)
(34)			MCDWB	3.6	8.1	14.0	18.8	22.8	28.6	31.5	30.9	27.8	20.4	12.2	5.5	(34)
(35)	Mean Daily Temperature Range	MDBR	8.7	10.6	12.0	12.2	13.2	14.7	14.3	14.5	15.0	13.4	11.9	8.5	(35)	
(36)		5% DB	MCDBR	9.8	13.2	14.7	15.4	15.3	16.1	15.3	15.8	16.8	16.0	14.2	10.3	(36)
(37)			MCWBR	6.5	7.8	7.4	6.8	5.8	5.5	5.3	5.5	6.4	7.3	8.4	7.1	(37)
(38)		5% WB	MCDWB	8.6	11.7	12.7	13.2	13.6	15.2	14.5	14.7	15.8	13.6	12.2	8.7	(38)
(39)			MCWBR	5.9	7.2	6.7	6.1	5.8	5.6	5.7	5.9	6.6	6.6	7.7	6.2	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.315	0.353	0.413	0.485	0.454	0.432	0.480	0.463	0.396	0.393	0.334	0.311	(40)	
(41)		taud	2.312	2.195	2.064	1.917	2.070	2.190	2.064	2.106	2.275	2.210	2.357	2.361	(41)	
(42)		Ebn at Noon	837	848	830	793	830	848	802	805	841	792	809	813	(42)	
(43)		Edh at Noon	90	117	150	186	163	145	163	152	120	115	86	80	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.33	3.35	3.97	4.65	5.71	6.93	6.95	6.37	5.20	3.41	2.30	1.75	(44)	
(45)		RadStd	0.25	0.42	0.38	0.53	0.33	0.42	0.31	0.32	0.31	0.25	0.31	0.24	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
(46)	Station Only	(d) +0.56	(e) N/A	(f) N/A	(g) +1.59	(h) -0.93	(i) -1.23	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(46)
(47)	Regional (14 neighbors)	N/A	N/A	N/A	+0.82	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page